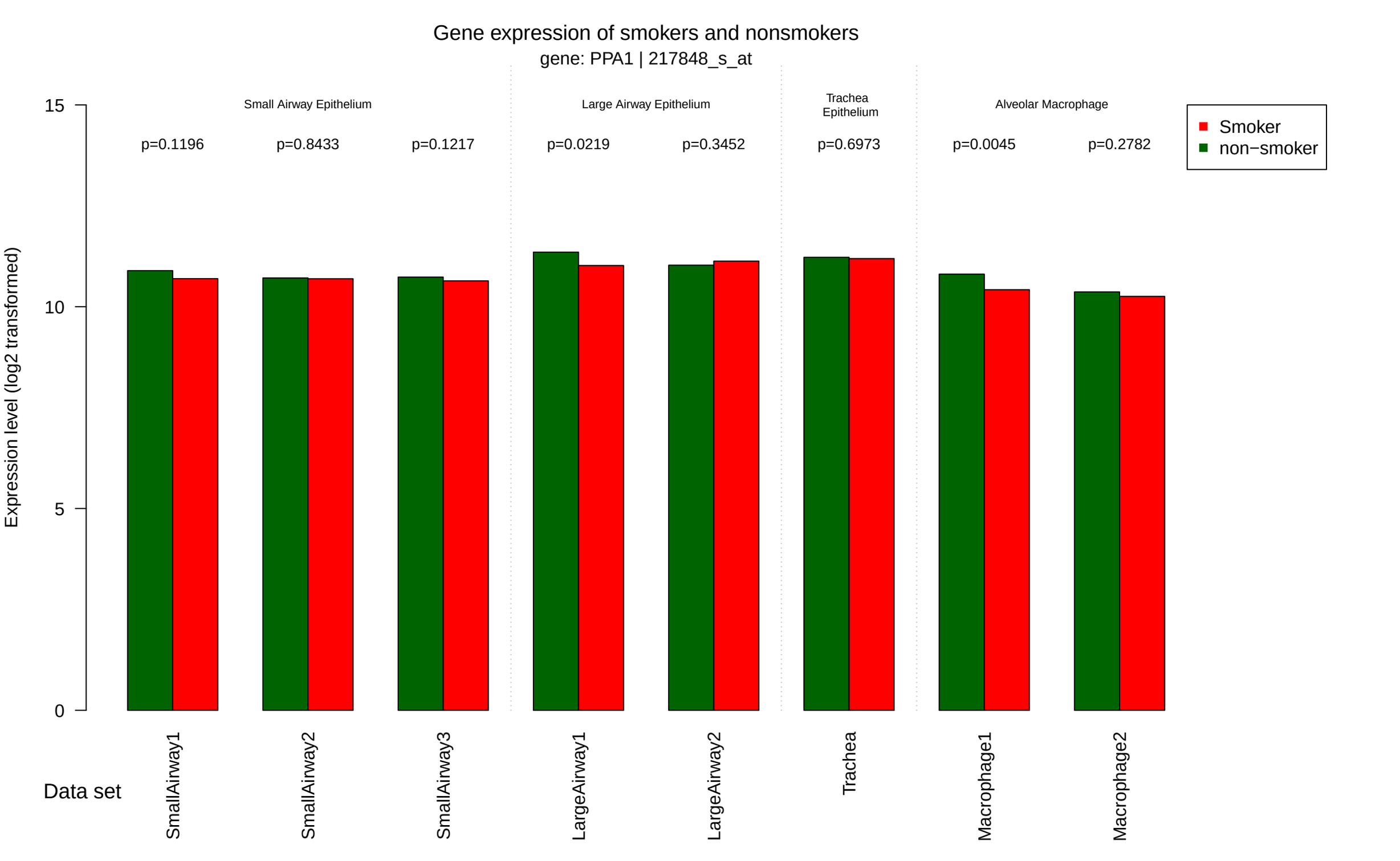
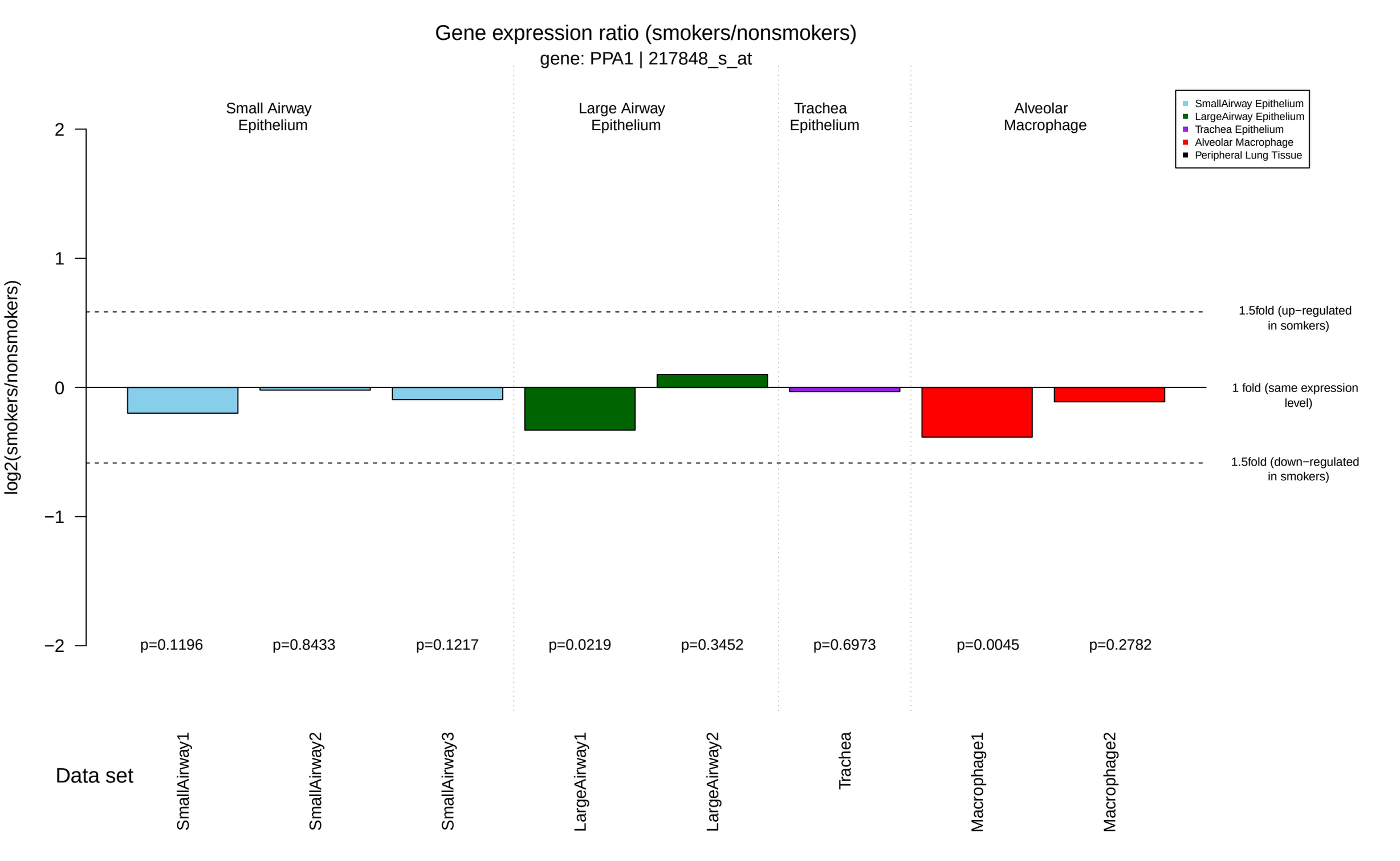
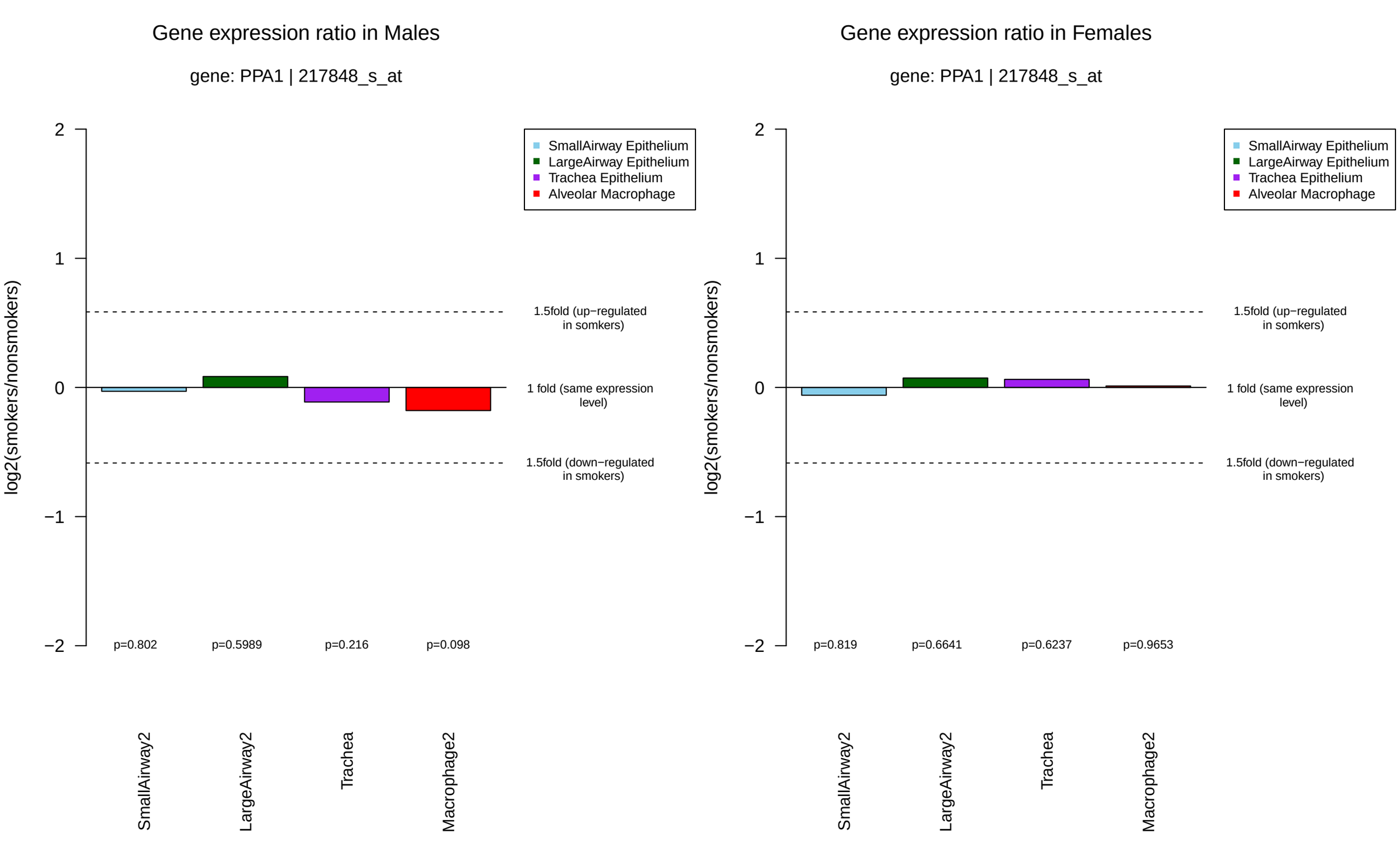
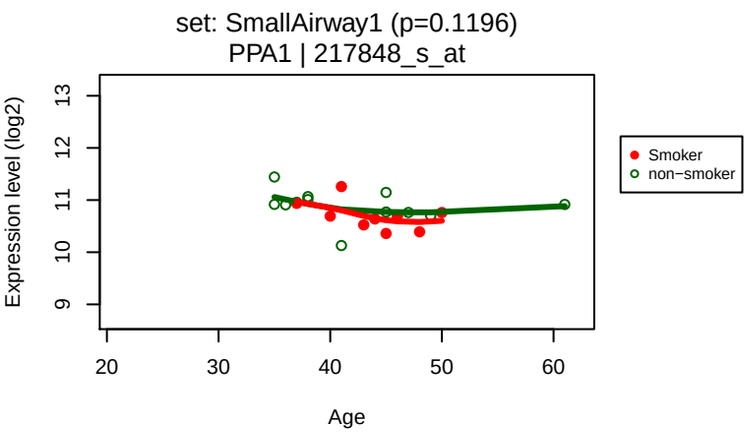
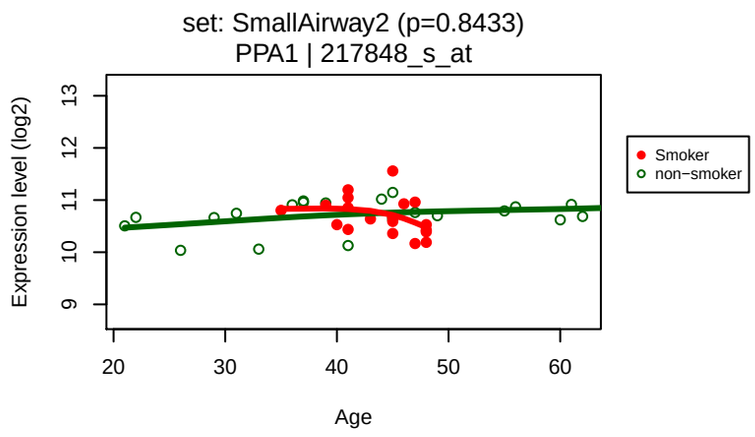
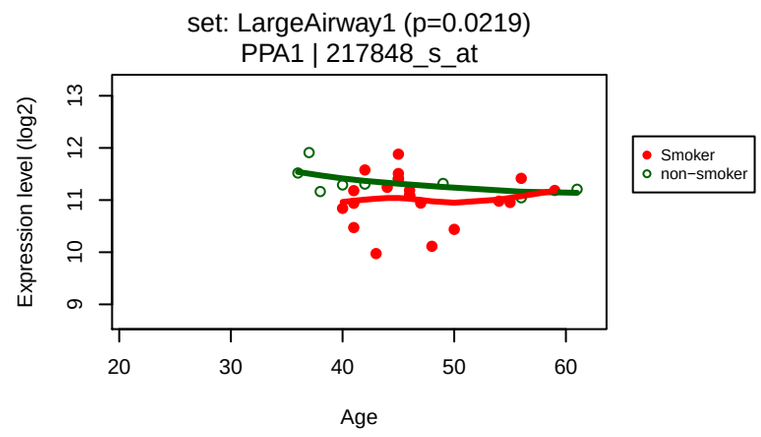

Gene expression of smokers and nonsmokers
gene: PPA1 | 217848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1196
p=0.8433
p=0.1217
p=0.0219
p=0.3452
p=0.6973
p=0.0045
p=0.2782
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

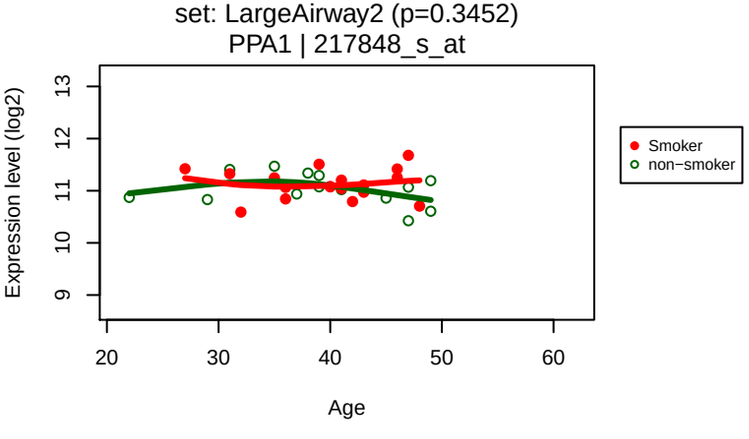

Gene expression ratio (smokers/nonsmokers)
gene: PPA1 | 217848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1196
p=0.8433
p=0.1217
p=0.0219
p=0.3452
p=0.6973
p=0.0045
p=0.2782
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

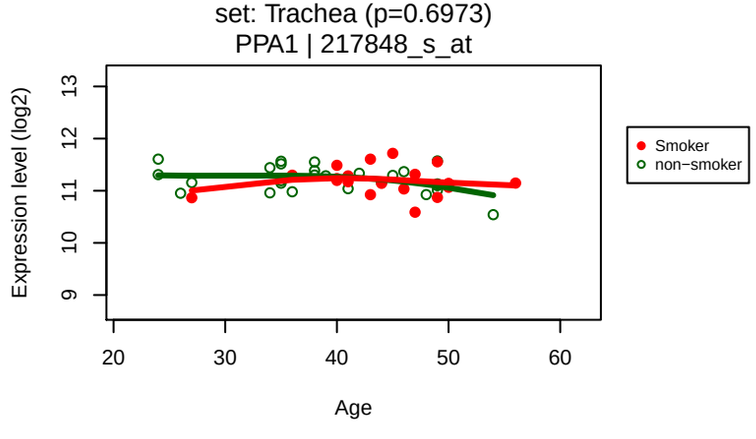

Gene expression ratio in Males
gene: PPA1 | 217848_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.802
p=0.5989
p=0.216
p=0.098
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPA1 | 217848_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.819
p=0.6641
p=0.6237
p=0.9653
SmallAirway2
LargeAirway2
Trachea
Macrophage2

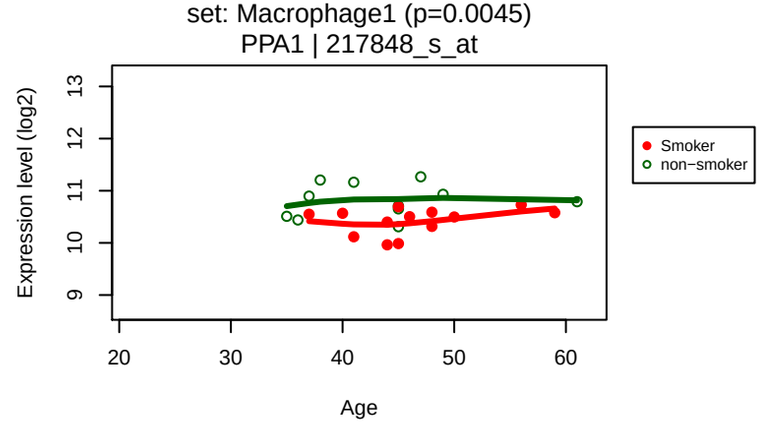

set: SmallAirway1 (p=0.1196)
PPA1 | 217848_s_at
Expression level (log2)
Age
Smoker
non−smoker

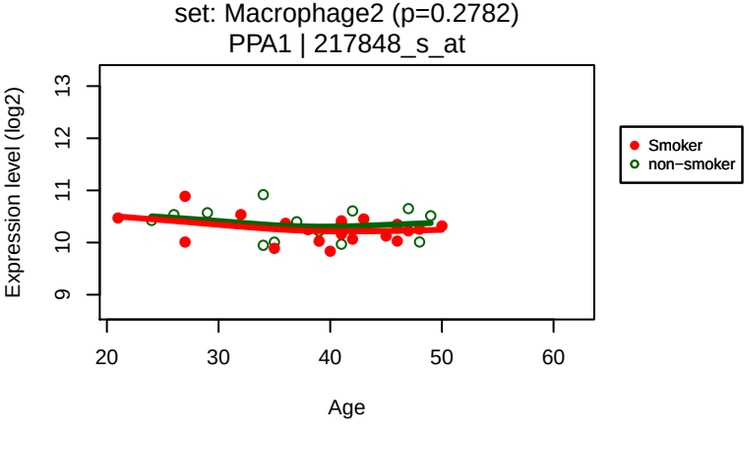

set: SmallAirway2 (p=0.8433)
PPA1 | 217848_s_at
Expression level (log2)
Age
Smoker
non−smoker

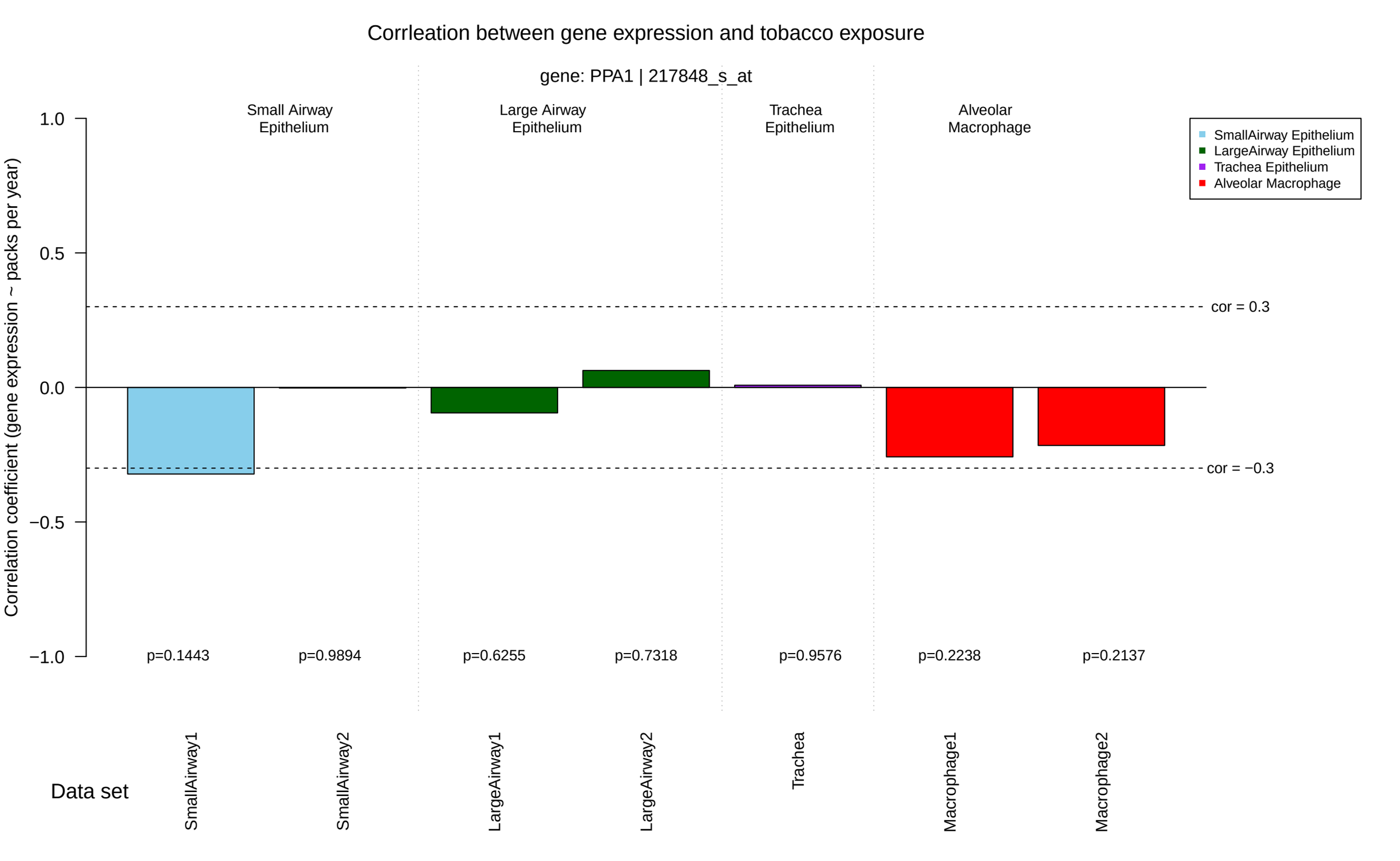

set: LargeAirway1 (p=0.0219)
PPA1 | 217848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3452)
PPA1 | 217848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6973)
PPA1 | 217848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0045)
PPA1 | 217848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2782)
PPA1 | 217848_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPA1 | 217848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1443
p=0.9894
p=0.6255
p=0.7318
p=0.9576
p=0.2238
p=0.2137
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2